INCH-POUND

MIL-DTL-17/182C
w/AMENDMENT 1
22 June 2019
SUPERSEDING
MIL-DTL-17/182C
25 July 2014

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, TWIN, 95 OHMS, M17/182-00001 UNARMORED, M17/182-00002 ARMORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

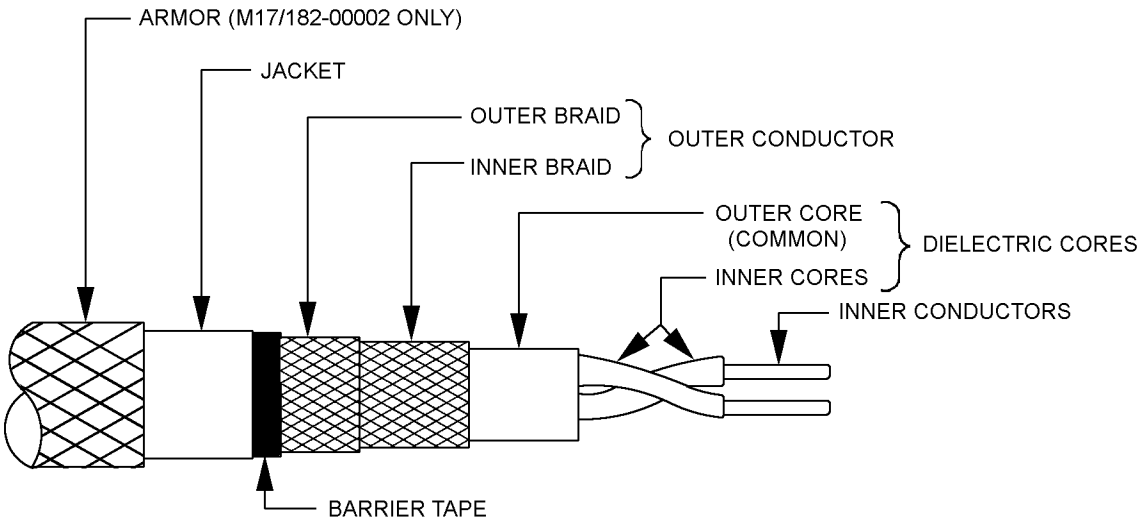


FIGURE 1. General configuration.

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TABLE I. Description.

Component	Construction
Inner conductors	Two conductors. Seven strands of bare copper wire, each strand .0152 inch each diameter. One strand of one conductor shall be tinned copper for identification. Overall diameter of each conductor: .0456 inch ±0.0010
Dielectric cores: Inner cores (individual) Outer core (common)	Two individual cores, under a common core. Two cores, twisted together with a right-hand lay of 4.25 ±0.5 inches. Type A-2: Solid, polyethylene, talc-coated each core. Diameter: .090 inch ±0.002 Type A-1: Solid, polyethylene, filled to round. Diameter: .285 inch ±0.006
Outer conductor: Inner braid Outer braid	Double braid of AWG No. 34, bare copper wire Diameter: .355 inch maximum Coverage: 94.1% nominal Carriers: 24 Ends: 8 Picks/inch: 9.1 ±10% Coverage: 96.0% nominal Carriers: 24 Ends: 8 Picks/inch: 12.0 ±10%
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape shall be applied with a 50% lap, aluminum face toward the outer conductor. Diameter: .365 inch maximum
Jacket	Type XIV, Cross-linked polyolefin Diameter: .420 inch ±0.010
Armor (M17/182-00002 only)	Single braid of aluminum-alloy wire Diameter: .490 inch maximum

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ENGINEERING INFORMATION

Continuous working voltage: 750 V rms, maximum.

Operating frequency: 200 MHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -30°C to +85°C.

Inner conductor properties:

DC resistance (maximum at +20°C): .68 ohms per 100 feet (each conductor).

Elongation: 20 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in balanced cables applications.

REQUIREMENTS (See table II)

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Diameter: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

(($T_{MAX} - T_{MIN}$) shall be interpreted as the difference in total wall thickness measured (1) along a line through the centers of the conductors (2) between a line tangent to the two conductors and a parallel line tangent to the surface of the core on both sides of the conductors. The eccentricity corresponding to the larger measured value ($T_{MAX} - T_{MIN}$) shall be reported.

Adhesion of conductors:

Inner conductor to core: 2 pounds, minimum; 20 pounds, maximum.

Electrical and mechanical:

Continuity: Applicable.

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 2,000 V rms, minimum.

Insulation resistance: Not applicable.

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Corona extinction voltage: Not applicable.

Characteristic impedance: 95 ohms ± 5.

RF transmission loss (attenuation): 4 db per 100 feet, maximum at 100 Mhz.
6 db per 100 feet, maximum at 200 Mhz.

Standing wave ratio (return loss): Not applicable.

Capacitance: 17.4 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: 5 percent, maximum.

Transmission unbalance: 10 percent, maximum, on a 100 foot length when measured between 100 and 160 Mhz.

Inductance: Not applicable
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable

Aging stability: +98ºC ± 2ºC.

Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.

Cold bend: -30°C ± 2°C.

Dimensional stability: +85ºC ± 2ºC.

Inner conductor from core: .062 inch, maximum.

Inner conductor from jacket: .125 inch, maximum.

Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.

Flame propagation: Applicable.

Acid gas generation: 2 percent, maximum.

Halogen content: 0.2 percent %, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

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Weathering: Applicable.

Abrasion resistance: 75 cycles minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation: 160 percent, minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongation, percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight:

14.2 pounds per 100 feet, maximum. (M17/182-00001)

16.9 pounds per 100 feet, maximum. (M17/182-00002)

Marking: Applicable.

Workmanship: Applicable.

Part or Identifying Number (PIN):

M17/182-00001	UNARMORED
M17/182-00002	ARMORED

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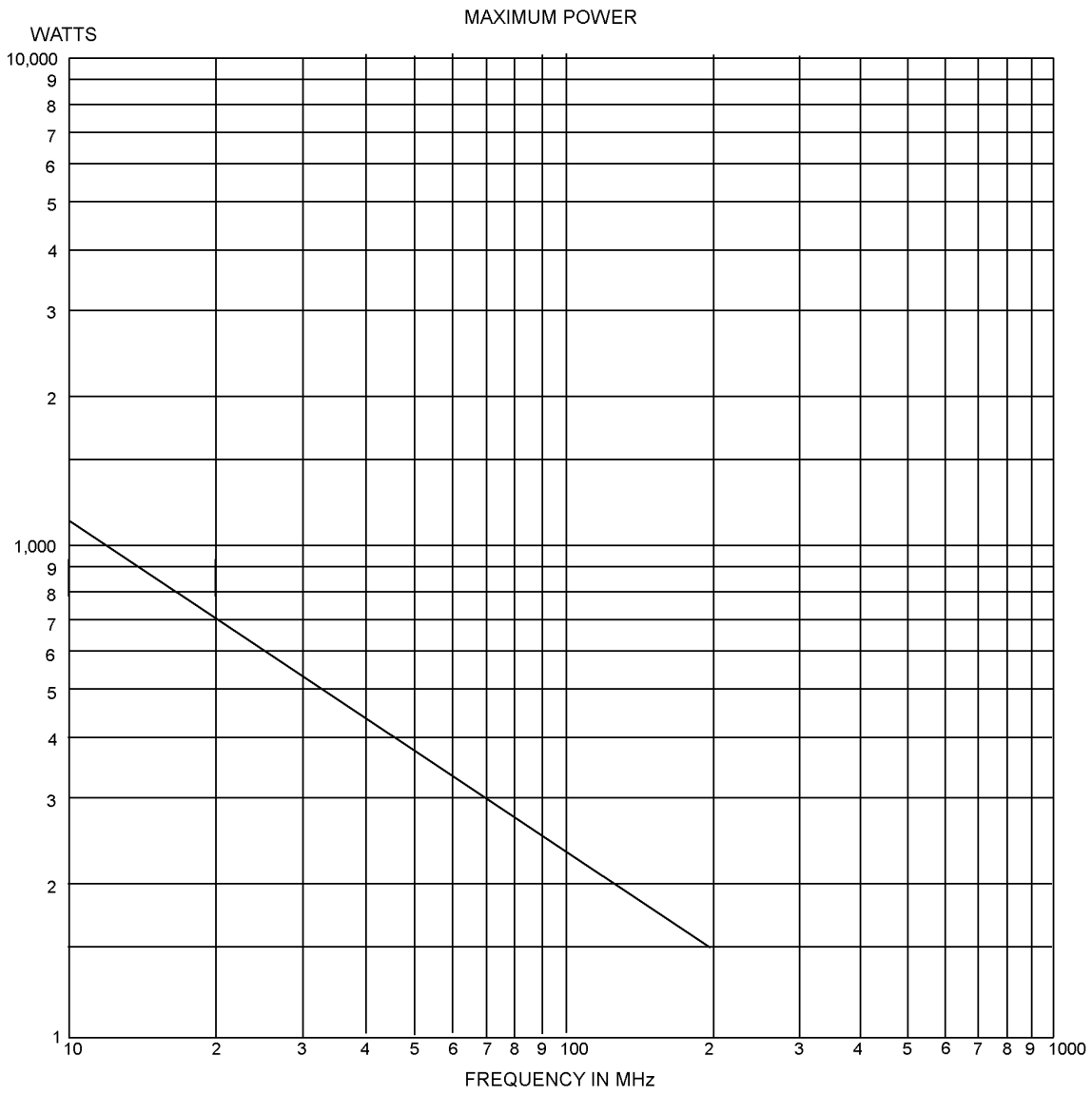


FIGURE 2. Power rating at sea level and 25ºC.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	3.7.11	4.8.12	1	Per Table VIII	B
	Transmission unbalance	3.7.12	4.8.13	1	Per Table VIII	B
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				

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TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C
	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion tests	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity Index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals (aged)	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity
DLA - CC

(Project 6145-2019-019)

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